BatterySense AI Battery Cycling Analysis Report

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Dataset summary

Rows	856
Columns	12
Uploaded files	cell_01.csv, cell_02.csv, cell_03.csv
Cells/samples	3
Anomaly flags	12
Excluded rows	5
Best cell	cell_01
Worst cell	cell_03

Key metrics

cell_or_sample	n_points	first_charge_or_lit...	first_discharge_or_...	first_coulombic_eff...	max_discharge_or_d...	last_retention_capa...	last_capacity_reten...	last_capacity_reten...	degradation_rate_pc...	average_coulombic_l...	last_accumulated_ce...	total_accumulated...
cell_01	323	206.4	178.7	86.57	202	192.3	107.6	95.21	-0.004593	99.84	99.84	101
cell_02	210	290.9	243.4	83.69	243.4	184.4	75.73	79.89	-0.07066	99.7	99.66	144.9
cell_03	318	291.1	262.6	90.23	262.6	197.2	75.07	77.41	-0.04067	99.68	99.69	234.8

Multi-cell deviation / reproducibility report

parameter	n_cells	mean	std	rsd_pct	min	max	range
STD in first-cycl...	3	86.83	3.276	3.773	83.69	90.23	6.536
STD in average CE...	3	99.74	0.0902	0.09044	99.68	99.84	0.1653
STD in first-cycl...	3	262.8	48.82	18.58	206.4	291.1	84.66
STD in first-cycl...	3	228.3	43.99	19.27	178.7	262.6	83.94
STD in upper/max ...	3	262.8	48.82	18.58	206.4	291.1	84.66
STD in upper/max ...	3	236	31	13.14	202	262.6	60.66

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parameter	n_cells	mean	std	rsd_pct	min	max	range
STD in final rete...	3	84.17	9.641	11.45	77.41	95.21	17.8
STD in degradatio...	3	-0.03864	0.03308	-85.61	-0.07066	-0.004593	0.06607

Per-cell deviation from mean

cell_or_sample	first_coulombic_efficiency_pct	average_coulombic_efficiency_pct	first_charge_or_lithiation_capacity	first_discharge_or_delithiation_capacity	max_charge_or_lithiation_capacity	max_discharge_or_delithiation_capacity	last_capacity_retention_pct
cell_01	86.57	99.84	206.4	178.7	206.4	202	95.21
cell_02	83.69	99.7	290.9	243.4	290.9	243.4	79.89
cell_03	90.23	99.68	291.1	262.6	291.1	262.6	77.41

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Scientific interpretation

Scientific interpretation

Backend status: Rule-based local analysis was selected. No API or local LLM was used.

Executive summary

The BatterySense AI local analysis identified **3 cell/sample group(s)**. Based on the max-stabilized capacity-retention baseline, the best-performing cell/sample is **cell_01**, while the weakest cell/sample is **cell_03**. The mean final retention is **84.17%**, the mean first-baseline retention is **86.14%**, and the mean calculated Coulombic efficiency is **99.74%**.

Data and protocol overview

The uploaded data were cleaned, standardized, and analyzed locally. CE and related quantities were calculated from the mapped charge/lithiation and discharge/delithiation capacities using the selected cell-type convention. The dataset includes the following protocol-segment counts: **{'cycling': 806, 'rate_test': 47, 'formation': 3}**. Excluded rows after quality filtering: **5**.

Formation and early-cycle behaviour

First-cycle charge/lithiation capacity, discharge/delithiation capacity, and first-cycle CE are important indicators of irreversible losses during initial conditioning. In this analysis, the first-cycle and upper-capacity values are reported per cell so that formation reproducibility can be assessed directly.

• **cell_01**: first charge/lithiation 206.42, first discharge/delithiation 178.69, first CE 86.57%, max discharge/delithiation 201.98, last capacity 192.30, retention from first 107.62%, retention from max-stabilized 95.21%, average CE 99.842%, degradation slope -0.0046% per cycle.

• **cell_02**: first charge/lithiation 290.89, first discharge/delithiation 243.45, first CE 83.69%, max discharge/delithiation 243.45, last capacity 184.36, retention from first 75.73%, retention from max-stabilized 79.89%, average CE 99.697%, degradation slope -0.0707% per cycle.

• **cell_03**: first charge/lithiation 291.08, first discharge/delithiation 262.63, first CE 90.23%, max discharge/delithiation 262.63, last capacity 197.15, retention from first 75.07%, retention from max-stabilized 77.41%, average CE 99.677%, degradation slope -0.0407% per cycle.

Capacity evolution and SOH

BatterySense AI reports two capacity-retention baselines. The first-valid baseline shows change relative to the first valid output capacity, while the max-stabilized baseline compares each later point with the maximum stable capacity observed during cycling. The max-stabilized baseline is used for the main SOH and final ranking because it is often more representative after formation stabilization.

Coulombic efficiency and irreversible capacity

Average CE, accumulated CE, CE_20_pt_AVG, and accumulated irreversible capacity should be interpreted together. Stable CE near 100% generally indicates good reversibility, whereas low or unstable CE can indicate irreversible side reactions, wetting/formation issues, or measurement instability. These mechanistic explanations remain tentative without metadata such as temperature, electrolyte, loading, and protocol details.

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Cell-to-cell comparison and reproducibility

The deviation table quantifies reproducibility using STD and RSD across cells. Lower STD/RSD in first-cycle CE, first-cycle capacities, upper/max capacities, final retention, and degradation slope indicates more reproducible cell behaviour.

- **STD in first-cycle CE (%)**: mean 86.828, STD 3.276, RSD 3.77%, range 6.536.
- **STD in average CE (%)**: mean 99.739, STD 0.090, RSD 0.09%, range 0.165.
- **STD in first-cycle charge/lithiation capacity**: mean 262.798, STD 48.821, RSD 18.58%, range 84.656.
- **STD in first-cycle discharge/delithiation capacity**: mean 228.259, STD 43.985, RSD 19.27%, range 83.943.
- **STD in upper/max charge/lithiation capacity**: mean 262.798, STD 48.821, RSD 18.58%, range 84.656.
- **STD in upper/max discharge/delithiation capacity**: mean 236.021, STD 31.003, RSD 13.14%, range 60.657.
- **STD in final retention (%)**: mean 84.168, STD 9.641, RSD 11.45%, range 17.802.
- **STD in degradation rate (%/cycle)**: mean -0.039, STD 0.033, RSD -85.61%, range 0.066.

Anomaly and data-quality assessment

BatterySense AI detected **12** anomaly flag(s). These should be treated as diagnostic flags rather than final failure assignments. Confirm flagged points against the original tester files and channel logs before making final conclusions.

Scientific interpretation

The observed differences between cells should be interpreted primarily from retention, degradation slope, CE stability, and reproducibility metrics. Large first-cycle capacity or CE differences suggest formation or electrode/process variability, while large divergence in final retention or degradation slope indicates differences in long-term cycling stability.

Recommended next experiments

1. Repeat or extend cycling for the best and worst cells to confirm reproducibility.
2. Inspect early-cycle CE and irreversible capacity to separate formation losses from long-term degradation.
3. Compare electrode loading, electrolyte amount, rest steps, current density, and temperature history for cells with poorer retention.
4. If available, add CC/CV split, dQ/dV, impedance, and metadata to improve mechanistic interpretation.
5. Validate anomaly flags against original tester logs before excluding or diagnosing a cell.

Final conclusion

The report provides an evidence-based ranking and quality-control screen. The strongest conclusion is the relative performance ranking from the locally calculated metrics; mechanistic explanations should be confirmed with protocol metadata and follow-up diagnostics.

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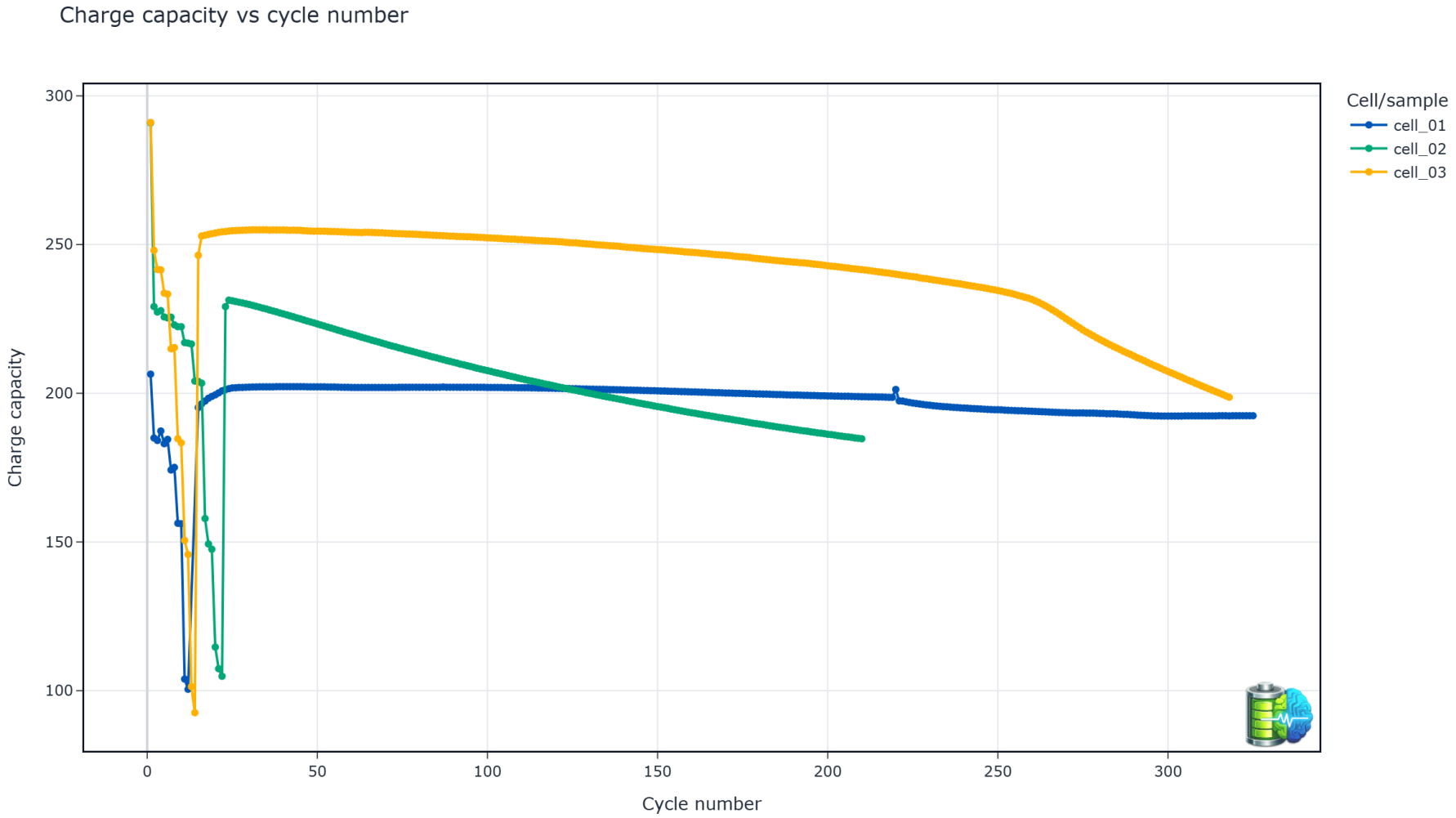
Static figure snapshots

Charge Capacity Vs Cycle



Discharge Capacity Vs Cycle

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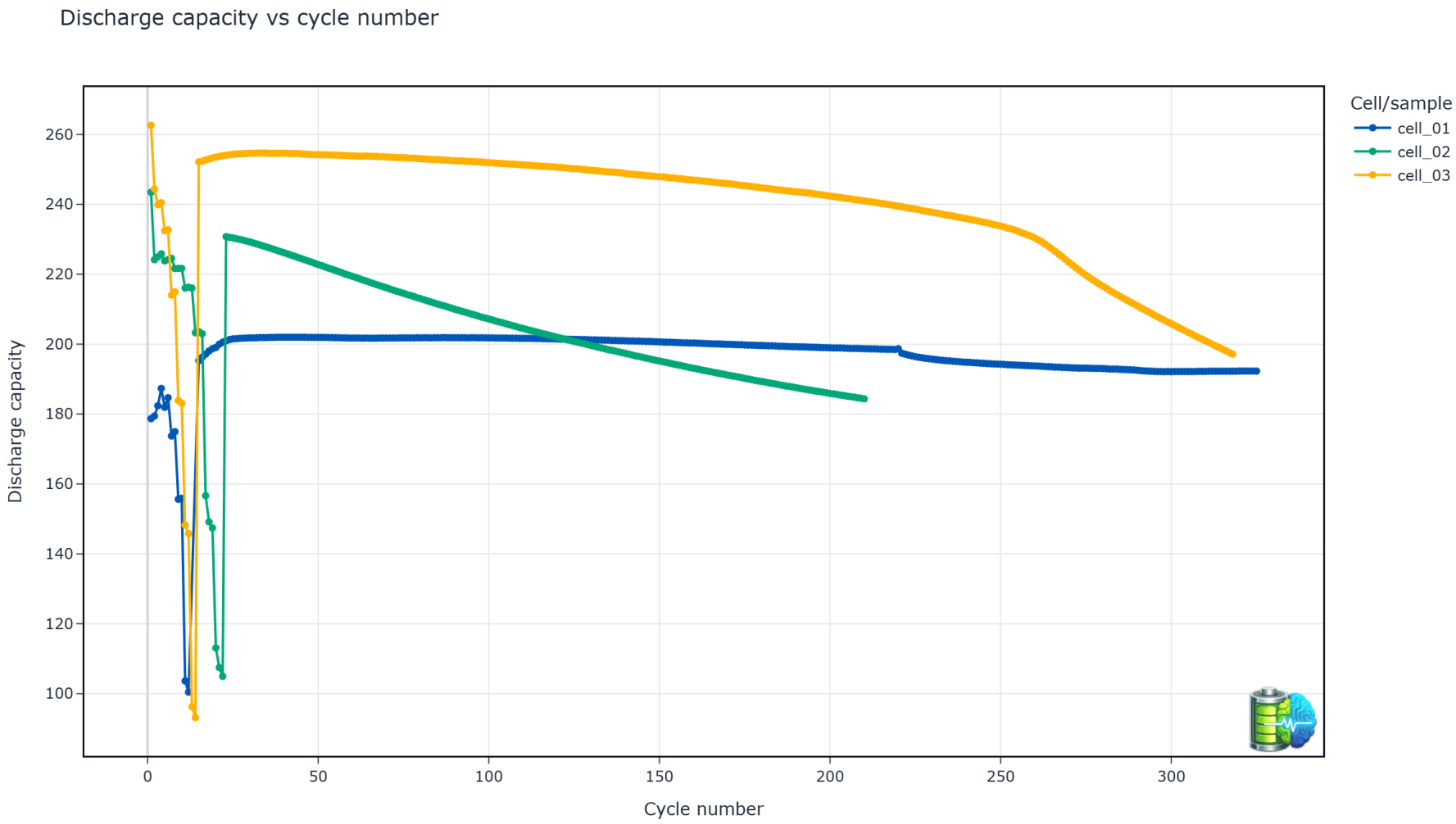

Coulombic Efficiency Vs Cycle

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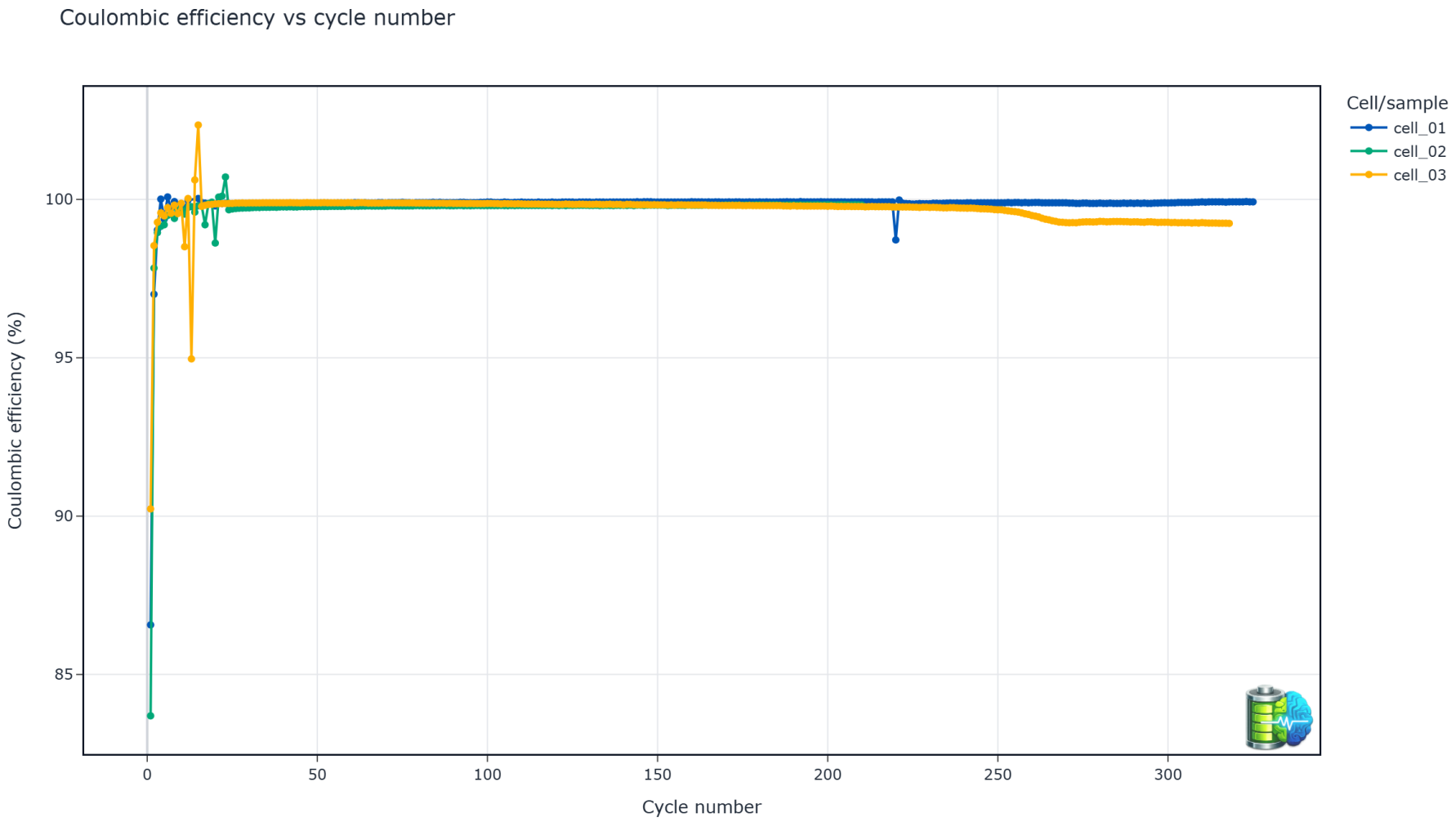

Capacity Retention Vs Cycle

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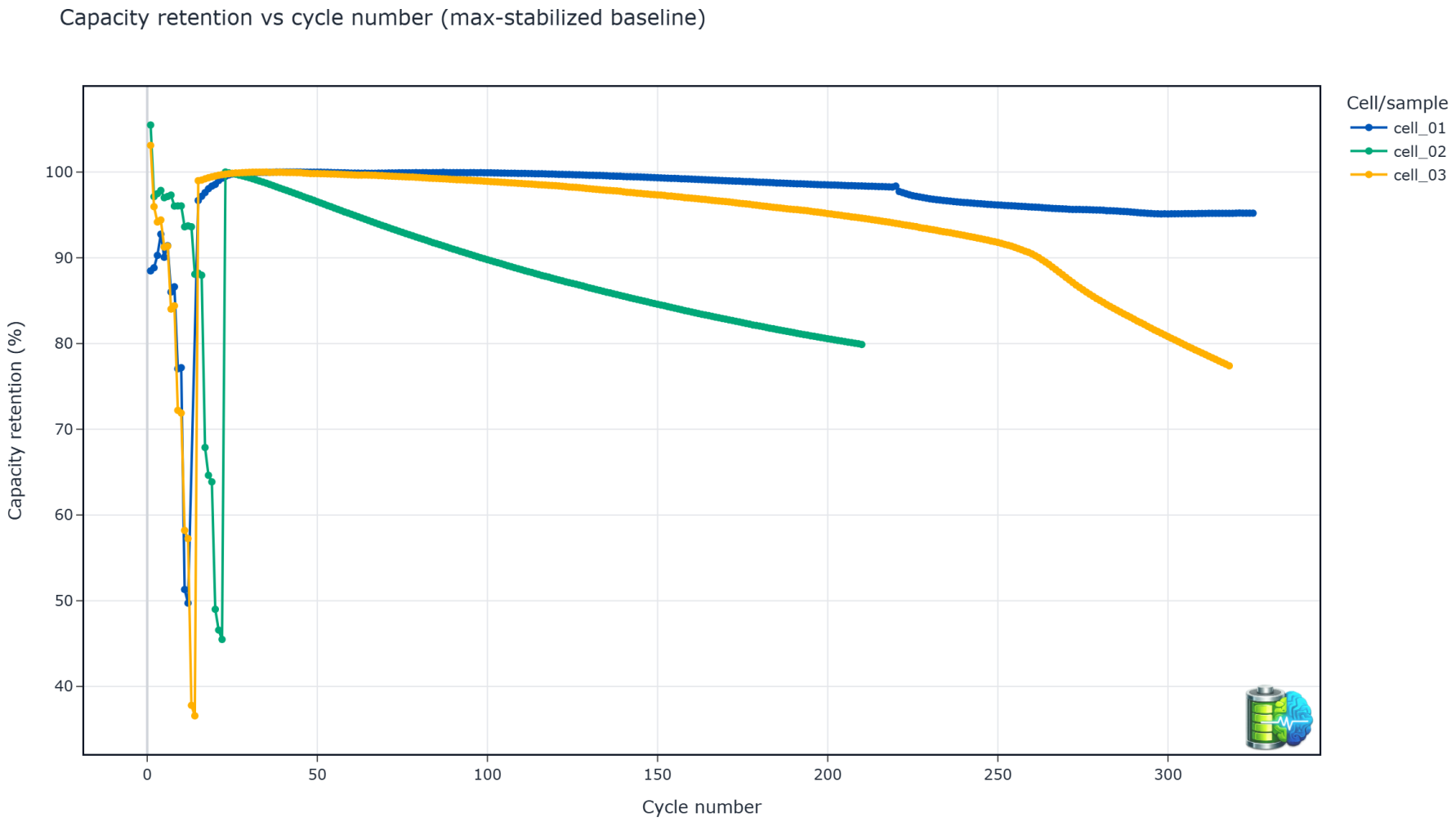

Capacity Retention Baseline Comparison

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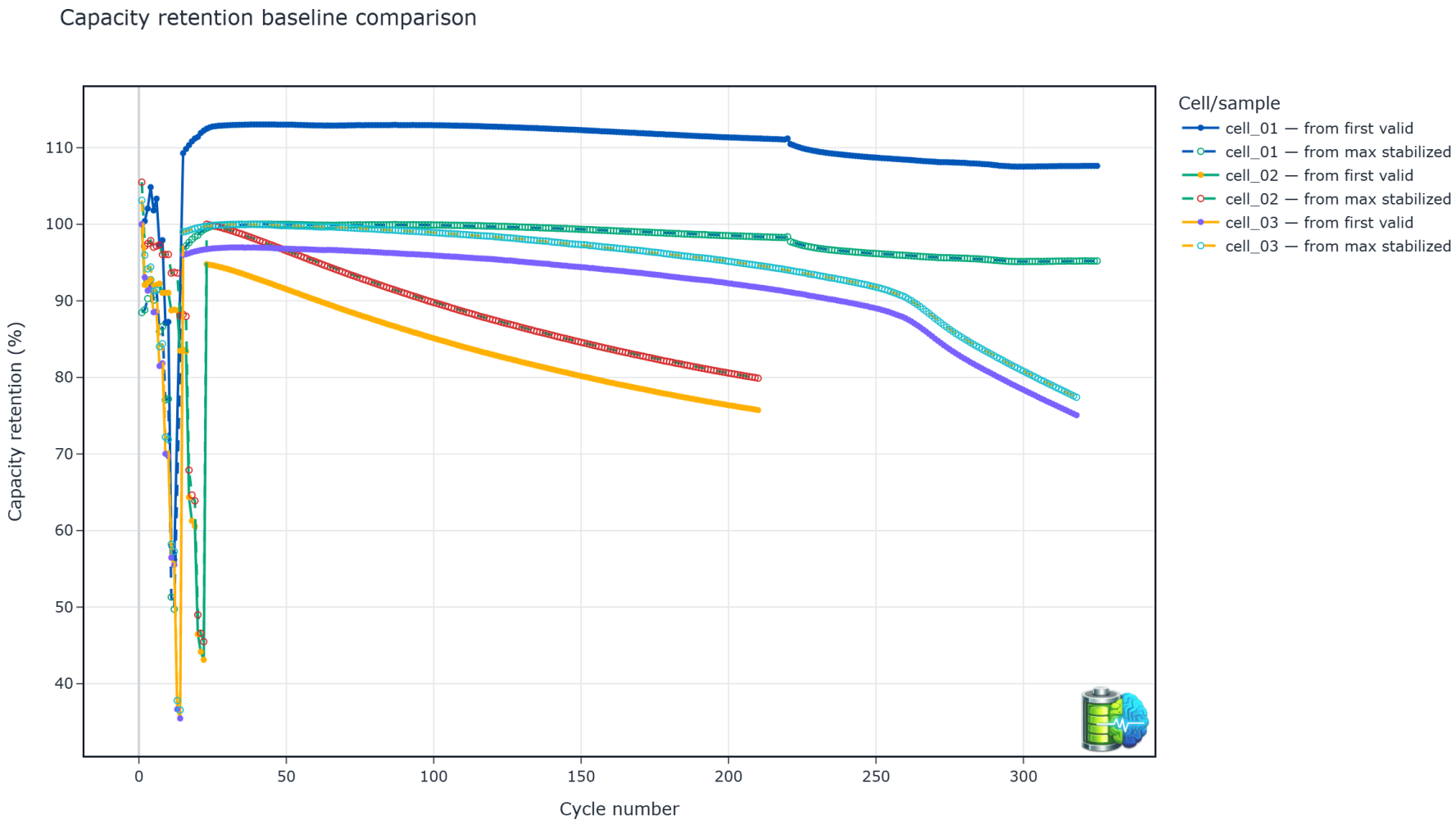

Ce 20 Point Average Vs Cycle

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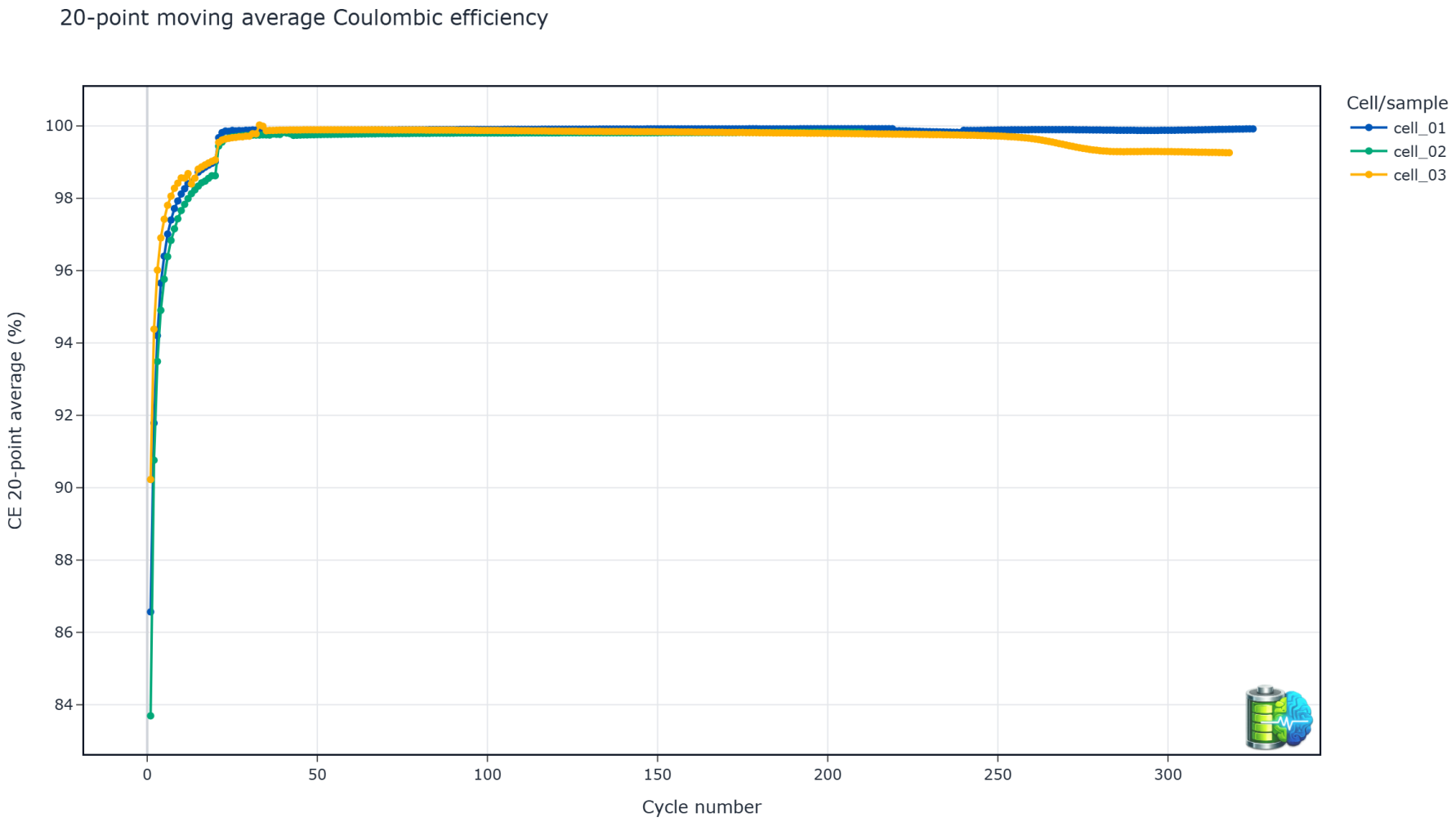

Accumulated Irreversible Capacity Vs Cycle

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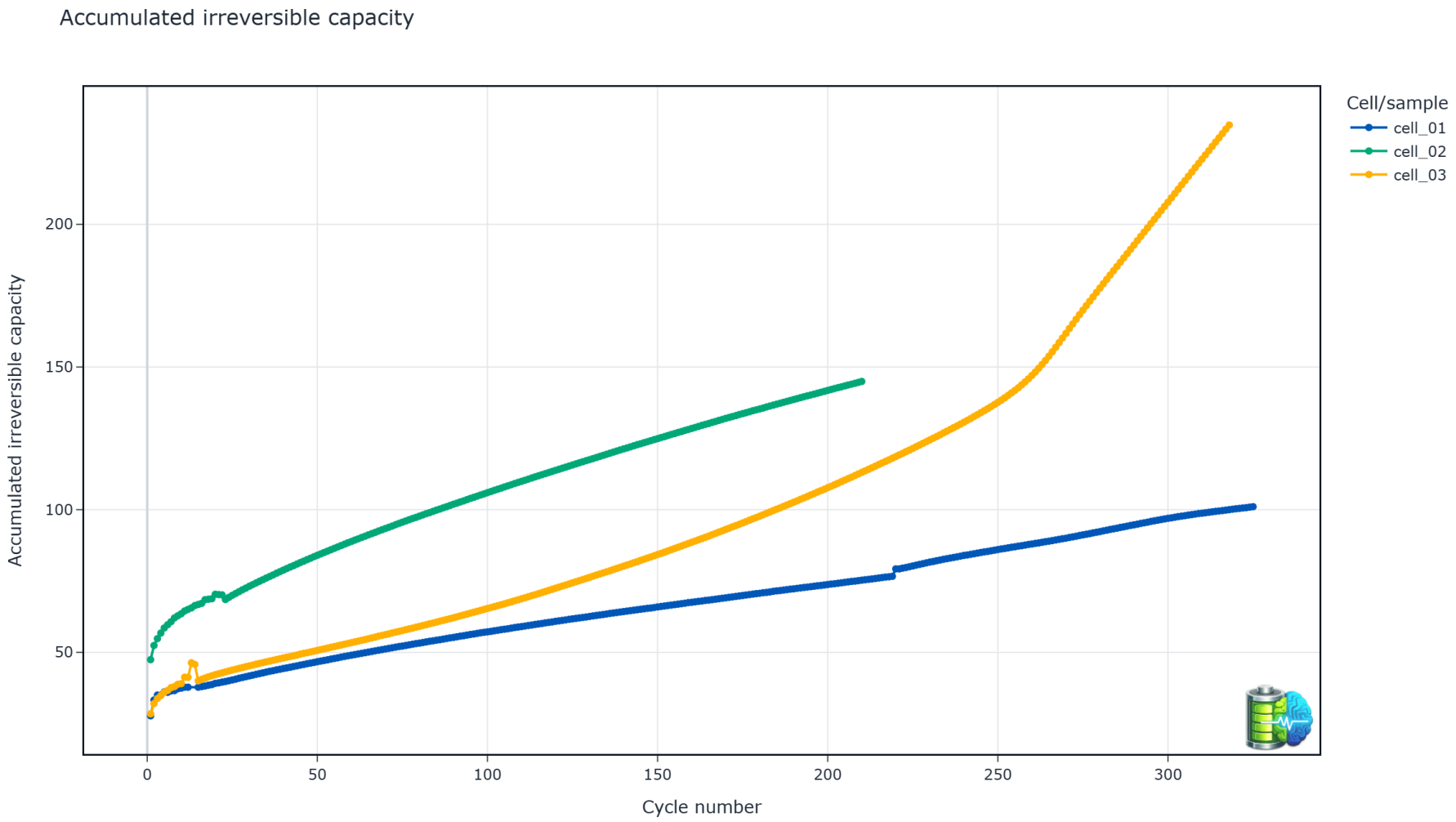

Protocol Segment Detection

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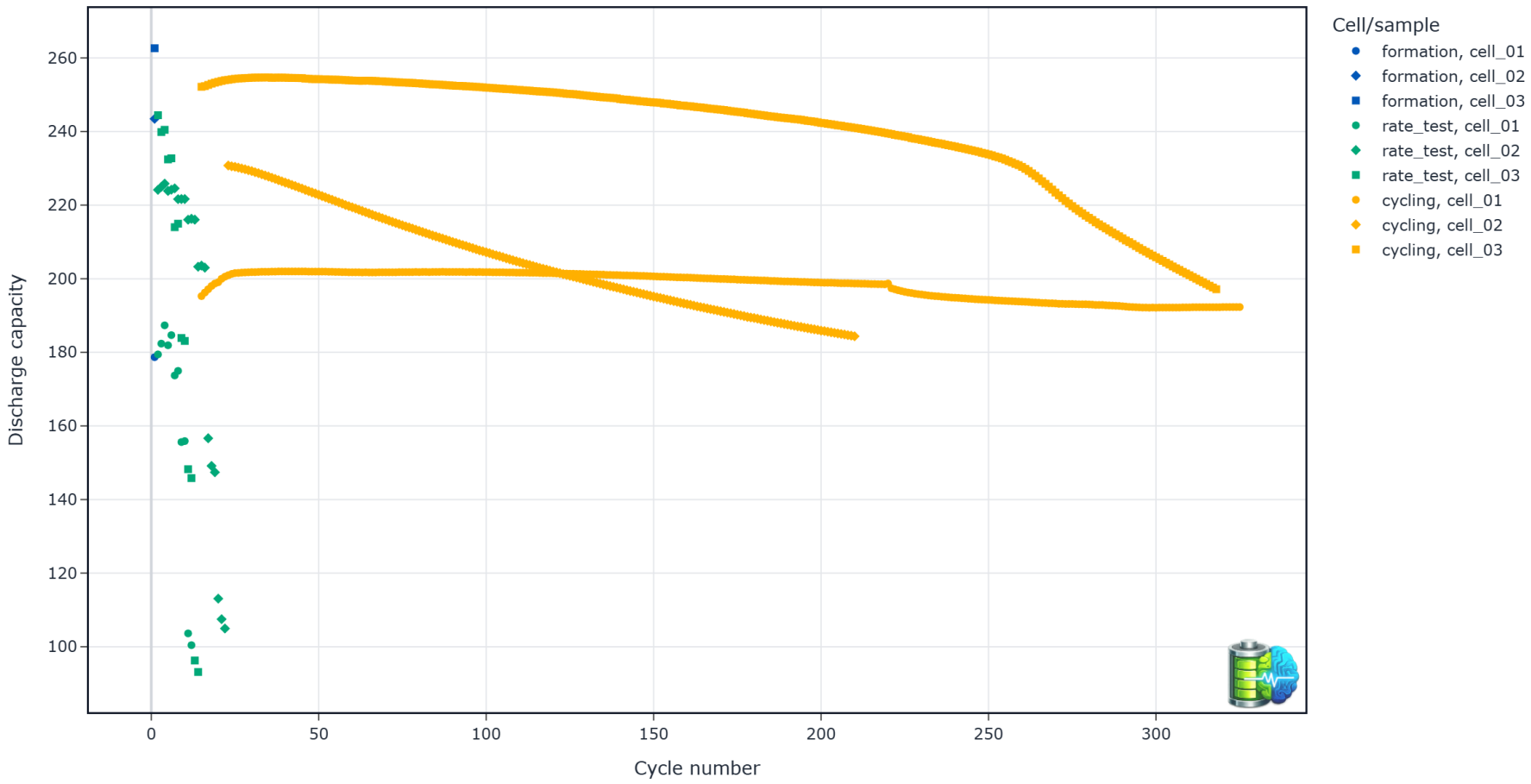
Detected protocol segments



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